



Aberdeen *Group*

The SOA in IT Benchmark Report

What CIOs Should Know about How SOA Is Changing IT

December 2005

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Executive Summary

The promise of service-oriented architecture (SOA) is real, and so is the CIO's dilemma. Although SOA offers such a compelling case for adoption, SOA's business benefits are still largely untapped as long as SOA activity sits only in the realm of IT.

Companies cannot expect business benefits from SOA before the IT department has been able to assimilate what SOA can do for them. The reality for IT executives and CIOs is that the IT benefits from SOA must precede business benefits, which are a cascading effect of full-lifecycle SOA implementations.

This dilemma is real, and companies must address it by recognizing that SOA holds — first and foremost — a technical promise of IT cost reduction as a result of faster and easier customization, integration, and deployment of IT capabilities. The causal relationship for the business is very simple: a more competitive business environment that is able to change and react to market and customer needs without being bound by IT limitations.

SOA is here, but not widely distributed. Only 16% of companies that responded to a recent Aberdeen survey have had more than 24 months of experience with SOA (21% in large companies), and 15% of respondents have managed or completed at least three SOA-related projects (large companies: 27%). The larger the SOA footprint becomes, the more tangible the benefits it will yield.

Key Business Value Findings

There is an urgent need for CIOs to more fully embrace SOA as a strategic blueprint for the IT organization. The top challenge to SOA adoption, cited by 41% of survey respondents, is “limited visibility for SOA value.” This may be a golden opportunity for CIOs to focus on the end-result of SOA business benefits rather than become enamored by the technology itself. A compelling business context is much easier to sell once a well-defined SOA implementation is fully delivered.

Aberdeen is bullish on SOA. We predict that over the next five years, the top Global 2000 organizations have a potential to save \$53 billion from their IT budgets as a result of SOA bearing fruit in reducing software implementation costs. This doesn't include the cascading effects of enabling the business to become more responsive to change because IT can deliver on these changing requirements in a shorter timeframe.

Savings from SOA implementations result in two major undertakings: (1) reinvestment in new IT capabilities and (2) direct business impact from SOA enablement to generate new revenue or enable new business capabilities. It's no surprise that the top factor for implementing SOA, cited by 50% of survey respondents, is “development of new capabilities.” Another related factor is the impact SOA is having on IT budgets relating to innovation and software maintenance costs. On average, best-in-class (BIC) companies spend 29.6% of their IT budgets on innovation versus 18.5% for all companies surveyed. Furthermore, BIC companies report that they spend only 12.4% of their IT budgets on software maintenance as opposed to 27.3% for all participants.



Companies that have had experience in the full-cycle of SOA implementations have been able to quantify the benefits of their undertakings in three major categories: (1) speed of deployment, (2) easier integration, and (3) faster customization and updates. These categories carry cost- and time-savings elements that will spread to the rest of the business.

SOA is for the long term. Companies will try to balance the initial cost of developing SOA applications against the expected savings from the deployment, integration, testing, and maintenance phases. The savings are real in productivity gains, time, and costs. Whether the SOA development is done internally or outsourced, IT maintenance costs will fall. While the cost of running existing applications is low, the cost of maintaining them is high. This bodes well for justifying an SOA implementation.

Implications & Analysis

Marrying the business requirements with IT capabilities through the use of business process configuration, orchestration, and manipulation tools will become a new competency. This will contribute to a stronger alignment between IT and the business. Software deployment is moving from design/compile/run to assemble/configure/monitor. The right business services at the right granularity level can be very powerful catalysts for binding business language to IT capabilities.

By focusing on the architectural requirements behind SOA, companies can take advantage of its strategic value. The architecture part is what gives the services orientation its highest value. Sixty percent of BIC companies say they're re-engineering their software architectures, versus 32% of all companies surveyed. Yet, the role of the SOA architect is critical, but not enough. A company needs an SOA competency center to serve as a repository of best practices to efficiently master the effectiveness of SOA-related deployments.

Ultimately, the SOA promise is about giving IT a higher level of predictability in delivering IT, from technical capabilities to business value. Running IT like a service, adopting open service architectures, and linking business to IT organically via a business services orientation will help the IT organization achieve "SO-IT," or *Service-Oriented IT*.

Recommendations for Action

An SOA direction presents an opportunity for forward-thinking CIOs to flex their muscles. The savvy CIO will know how to capitalize on the SOA shift in order to funnel the savings into additional value-added activities. Fifty percent of respondents expect to be able to develop new business capabilities or new products and services and 43% expect to reuse applications via Web services. These key benefits are residual effects of SOA. Ultimately, IT wants to spend more on innovation and new capabilities. SOA adoption will lower the cost of IT and enable more responsiveness to business needs. The size of the SOA budget is at stake, and it must be increased to fully exploit the promise of SOA.

Because SOA is not widely deployed, it has not yet changed the IT delivery model. IT must change it to take full advantage of the SOA promise. SO-IT is about defining and delivering IT in terms of business services that line-of-business managers can understand. IT will need to decouple these business-level services from the IT-level services, but must be able to assemble them rapidly to deliver the required functionality.



Table of Contents

Executive Summary	i
Key Business Value Findings.....	i
Implications & Analysis	ii
Recommendations for Action	ii
<i>Chapter One: Issue at Hand</i>	1
Wide Adoption and Experience Remain Scarce	2
SOA Cost Savings Help the IT Budget	2
SOA Permanently Binds the Business with IT	3
SOA Enables the Development of New Capabilities	4
IT Benefits Precede Business Benefits	4
<i>Chapter Two: Key Business Value Findings</i>	6
Put the SOA Architect in Charge of Architecting, Not Project Management ..	6
The Role of the SOA Architect	7
CIOs Need to Raise the Visibility of SOA's Value	8
Experience with SOA Projects Matters	8
Retooling IT Skills: Business Analysts Are in Demand	9
<i>Chapter Three: Implications & Analysis</i>	11
How to Quantify SOA's Value	12
Key SOA Technologies	13
Adoption Rates Vary by SOA Technology	13
Pressures, Actions, Capabilities, Enablers (PACE).....	14
Areas of SOA Being Implemented	16
<i>Chapter Four: Recommendations for Action</i>	17
Recommended Actions for all Companies	17
Laggard Steps to Success	18
Industry Norm Step to Success	19
Best in Class Next Steps	19
Let SOA Provide Better IT Predictability	19
Featured Sponsors.....	20
Sponsor Directory	22



Table of Contents

Author Profile	23
<i>Appendix A: Research Methodology</i>	24
<i>Appendix B: Related Aberdeen Research & Tools</i>	26
About AberdeenGroup	27



Figures

Figure 1: Company Experience with SOA Projects	2
Figure 2: Top 3 Factors Driving SOA Implementations	4
Figure 3: Who Leads SOA Efforts in Your Organization?	7
Figure 4: Top Three Strategic Actions or Plans to Address/Implement SOA	9
Figure 5: Likely Scenarios in Three Years as a Result of SOA Adoption	10
Figure 6: Areas of SOA Being Implemented	11
Figure 7: The Three Dimensions of SOA's Value	12
Figure 8: SOA Technology Components Use	14
Figure 9: Areas of SOA Being Implemented or Planning to be Implemented	16

Tables

Table 1: Percent of Current and Planned SOA-Based Software Applications	3
Table 2: SOA Implementation Challenges and Responses	8
Table 3: Key SOA Technologies and Current Adoption	13
Table 4: PACE (Pressures, Actions, Capabilities, Enablers)	15
Table 5: Relationship between PACE and Competitive Framework	25

Chapter One: Issue at Hand

Key Takeaways

- The SOA promise is real, but the benefits have not been widely distributed.
- A \$10 billion company with a \$300 million IT budget can save \$30 million per year from a broad SOA adoption after a 5-year horizon of implementing SOA in at least 75% of its applications.
- CIOs need to increase the visibility and footprints of SOA projects rapidly to gain the maximum benefit.
- Timid SOA undertakings lower IT's credibility as an agent of change for the business.

Software deployment complexities and unpredictable delivery outcomes from long IT projects have been at the center of every debate on IT's value to the enterprise. Most IT organizations still spend a considerable portion of their budgets (on average, 64%) on developing, deploying, or integrating applications — a vexing situation for CIOs. This limits the amount of available resources that can be directed toward more visible outcomes, such as fueling innovation and creating new technological capabilities.

Enter the concept of service-oriented architecture (SOA), a two-year-old technology-driven construct that promises profound implications for IT and holds a real potential to change the IT delivery model deeply. SOA approaches cover many issues. This report focuses on the following questions:

- Is IT showing enough leadership to advance the SOA agenda? Or is it not taking enough risks and, therefore, not pushing far enough for breakthroughs?
- How can SOA bring IT and the business together?
- What must IT organizations do to fully exploit the potential advantages of SOA?

SOA offers a mix of lessons from the Internet and client-server phenomena. Like the Internet in its early days, SOA has a simple veneer, but a complex core. It's simultaneously a concept, a methodology, a set of technologies, and a range of business practices. And, in the same way the client-server architecture brought significant changes to the way software was developed and delivered, SOA holds another promise to yet again change the IT delivery model permanently.

Competitive Framework Key

The Aberdeen Competitive Framework defines enterprises as falling into one of the three following levels of practices and performance:

Laggards (30%) —practices that are significantly behind the average of the industry

Industry norm (50%) — practices that represent the average or norm

Best in class (20%) — practices that are the best currently being employed and significantly superior to the industry norm

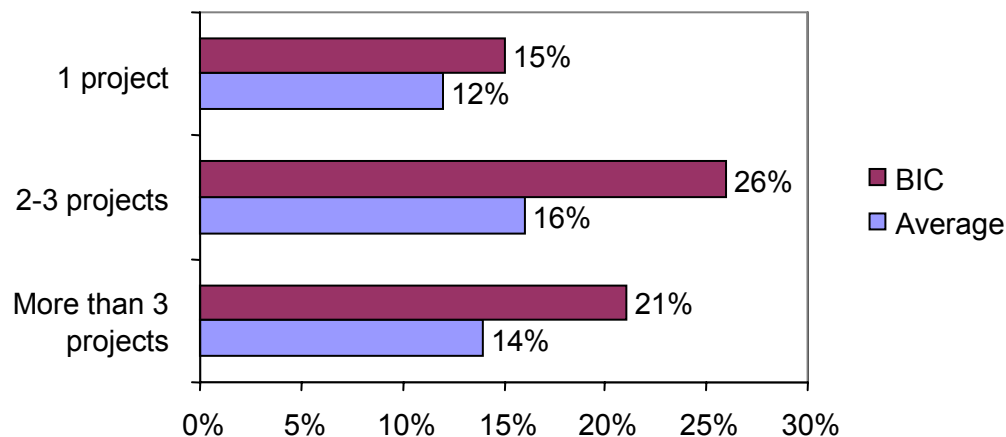


Wide Adoption and Experience Remain Scarce

Only 16% of companies in the Aberdeen survey have more than 24 months of experience with SOA technologies (21% among large companies). Twenty three percent (23%) have between 12 and 24 months of experience (large companies, 28%), and 22% have less than 12 months of experience (large companies, 29%). This indicates that experience and adoption levels vary widely across companies and tends to be higher for larger firms. (Figure 1).

As for personal experience with SOA projects, respondents at companies Aberdeen judges as best in class have more experience than average respondents have. This is not a surprise. Best-in-class companies have a leg up over others and are already reaping the benefits of SOA. The larger the SOA footprint becomes, the more tangible the benefits it will yield.

Figure 1: Company Experience with SOA Projects



Source: [AberdeenGroup](#), December 2005

SOA adoption levels are reminiscent of the early days of the Internet in 1995 and 1996, when even large companies had undertaken only two or three significant Internet projects. Although BIC respondents had more experience with SOA projects than the norm, SOA investments, in general, are still timid. One reason is that the psychology of IT spending is still oriented toward squeezing as much as possible from the IT budget. But SOA requires an investment approach to realize its benefits, especially in some of the infrastructure-related parts of its deployment. One IT executive from a large financial institution says it's difficult to take a pure ROI approach to early SOA investments. "It's a bit like paving the infrastructure. What's the ROI on pavement? But you need it to walk from one place to another."

SOA Cost Savings Help the IT Budget

Aberdeen is bullish on SOA. But only a widespread adoption of SOA technologies will provide enough of a visibly significant gain in productivity beyond the IT department. The first visible beachhead for which there are real SOA savings is related to software

applications. Based on survey analysis and interviews with several respondents, Aberdeen estimates that SOA adoption can save 25% of application deployment costs when a full lifecycle project is implemented. By coupling this estimate with survey data on respondents' predictions of usage over the next five years (Table 1), Aberdeen predicts

Over the next five years, Global 2000 companies will save up to \$53 billion from their IT budgets once SOA bears fruit.

that Global 2000 companies can potentially save \$53 billion from their IT budgets when SOA bears fruit in the form of reducing software implementation costs. This doesn't include the cascading effects of enabling the business to become more responsive to change, because IT can deliver on these changing requirements in a shorter timeframe.

Table 1: Percent of Current and Planned SOA-Based Software Applications

	None	1 to 10%	11 to 25%	26 to 50%	51 to 75%	76 to 100%
Today	32%	45%	14%	5%	2%	1%
In 2 years	2%	19%	38%	29%	8%	4%
In 5 years	0%	2%	12%	29%	37%	19%

Source: [AberdeenGroup](#), December 2005

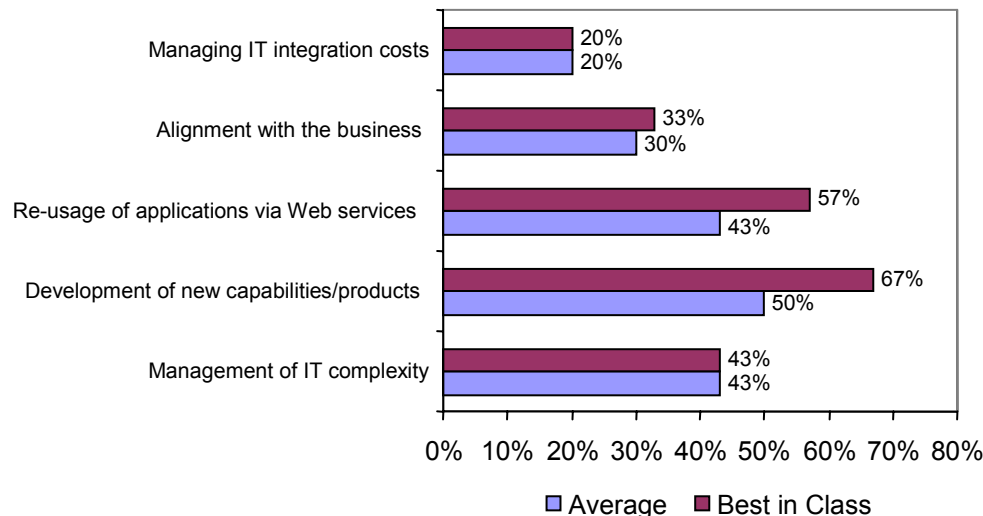
SOA Permanently Binds the Business with IT

SOA is a lever for IT, forcing a better business understanding of IT and enabling a new level of synchronization with the business. While technology and business changes are always expected, what has been missing is a tight linkage of the causal relationship between the two. Because SOA is grounded partly in IT and partly with the business, any change in one is automatically reflected in the other, in essence binding the two together permanently, and solving the age-old "IT alignment" challenge. SOA forces a joint, common thinking between the business and IT from the design to deployment phases.

It's no surprise that 50% of large companies cite "alignment with the business" as the top factor that drove their firms to initiate or implement SOA. This is closely followed by "managing IT complexity," also cited by 49% of large companies (Figure 2).



Figure 2: Top Three Factors Driving SOA Implementations



Source: [AberdeenGroup](#), December 2005

PACE Key — For more detailed description see Appendix A

Aberdeen applies a methodology to benchmark research that evaluates the business pressures, actions, capabilities, and enablers (PACE) that indicate corporate behavior in specific business processes. These terms are defined as follows:

Pressures — external forces that impact an organization's market position, competitiveness, or business operations

Actions — the strategic approaches an organization takes in response to industry pressures

Capabilities — the business process competencies required to execute corporate strategy

Enablers — the key functionality of technology solutions required to support the organization's enabling business practices

SOA Enables the Development of New Capabilities

Half of respondents to the Aberdeen survey are implementing SOA so they can take advantage of developing new business capabilities or new products and services. This is the case for 67% of best-in-class companies, further accentuating the importance of SOA as a catalyst that can unlock the potential for IT to contribute more directly to the business. In addition, 43% cited the re-use of applications via Web services as another key factor for SOA implementations.

IT Benefits Precede Business Benefits

The reality of SOA implementations is that IT benefits must precede business benefits. A software application based on an SOA infrastructure

looks the same on Day 1 of its rollout to a business user. However, it's the benefits obtained in the aftermath of the initial implementation that are most valuable for the business, namely in the speed of updates and required changes that are possible, i.e., how IT responds to the changes. It's wise to separate IT benefits from business accordingly:

SOA Benefits for IT:

- Management of IT complexity
- Re-use of applications
- Speed of IT implementations
- Lower IT integration costs

SOA Benefits for Business

- Faster reactions to change
- Competitiveness

Shared SOA Benefits for IT and Business

- Continuous alignment of IT and business
- Lower overall cost of IT
- Lower IT maintenance costs
- Lifecycle visibility into business process management
- Development of new IT-enabled capabilities



Chapter Two: Key Business Value Findings

Key Takeaways

- The role of the SOA architect is key, but not sufficient for managing SOA implementations.
- An SOA competency center is a critical component of SOA success.
- A company cannot achieve business value from an SOA until it realizes IT value from it first.
- There will be a surge in IT spending because of the need to upgrade the IT infrastructure to SOA.

SOA will ultimately change how IT capabilities are delivered. But there is no need to delay an IT organization's readiness to capitalize on such change. The key shift required is to enable IT's delivery as a set of business services, rather than as software technology. Most IT organizations today are organized according to existing software deployment practices, namely that software is designed or bought as a monolithic package. It's then customized, debugged, tested, and then deployed in various stages. Tomorrow's SOA-IT environment will enable the assembly of existing pieces as services. A business analyst will decide how these services will interact with each other via policy or capability configurations. An SOA integration specialist will lay out the orchestration and manipulation requirements for the services. Then, IT operations will monitor and manage the entire framework. Ultimately, business managers will be able to make changes to the software that powers their operations, but before we get there, IT must reach a certain level of SOA maturity that promises this plug-and-play environment at the business-level.

Who leads the IT organization through this transition is a critical aspect to becoming a *Service-Oriented IT* (SO-IT) organization.

Put the SOA Architect in Charge of Architecting, Not Project Management

By its very nature, implementing SOA requires coordination of multiple stakeholders within IT and across the organization. At a full maturity level, SOA leaves no part of the organization untouched. But at issue is what's required to get there. Who will lead the SOA strategy?

Forty-one percent (41%) of survey respondents indicate that the CIO is leading the SOA efforts within their organizations, and 40% say the IT architect/IT planning departments are the key leaders. This is not surprising. CIOs need to take a leadership position in SOA due to its strategic nature, and IT architects must lay out the blueprint for weaving services into the organization's IT fabric.

However, what's surprising in Figure 3 is that few companies have allowed a business manager to lead SOA efforts; even fewer are allowing an SOA competency center to take that role.

The reality is that a company needs an SOA architect to lay out the SOA vision, as well as an SOA competency center where best practices and SOA implementation experience are best leveraged for the rest of organization.

Before SOA can aspire to influence the business, it must first deliver on its technical promise; i.e., the deployment of fully operational end-to-end IT capabilities. The fact that few business managers are leading SOA efforts is a reflection of its real status today: SOA is still in the technical stage of its deployment.

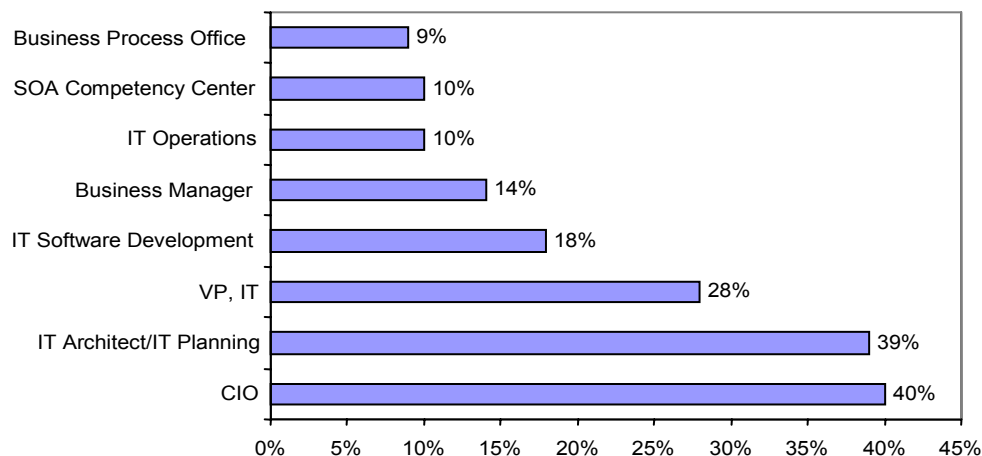
The Role of the SOA Architect

A key part of the SOA journey involves re-architecting the various SOA elements so services can be orchestrated appropriately. The role of the architect is emerging as a critical one, requiring a top-down approach to planning an SOA. Leading organizations are creating the position of “senior SOA architect,” a role responsible for the overall SOA blueprint, vision, governance, and transition roadmap. The SOA architect’s role is reminiscent of the re-engineering czar of the early 1990s or the internal Internet guru of the mid-1990s. The architect’s job requirements include a mix of technical, visionary, educational, evangelical, and change management skills. A key aspect of the SOA architect’s role is the working relationship with the application development group.

The importance of the SOA architect is primordial: 60% of BIC companies are re-engineering their software architecture versus 32% of all survey respondents. And, 64% of companies with more than 24 months of SOA experience predict there will be a surge in the prominence of IT architects, versus 48% of the entire survey field.

However, the SOA architect does not have to lead or manage SOA implementations. That task is best left to the SOA competency center or IT project office.

Figure 3: Who Leads SOA Efforts in Your Organization?



Source: [AberdeenGroup](#), December 2005



CIOs Need to Raise the Visibility of SOA's Value

Standing in the way of SOA implementations, according to 41% of survey respondents, is limited visibility for SOA value (Table 2). CIOs have a marketing role to play, but first they must deliver fully functional SOA applications to the business. Therefore, it's very important to frame the scope of the first SOA project in such a way that SOA's value can be visible both within IT and for the business.

Table 2: SOA Implementation Challenges and Responses

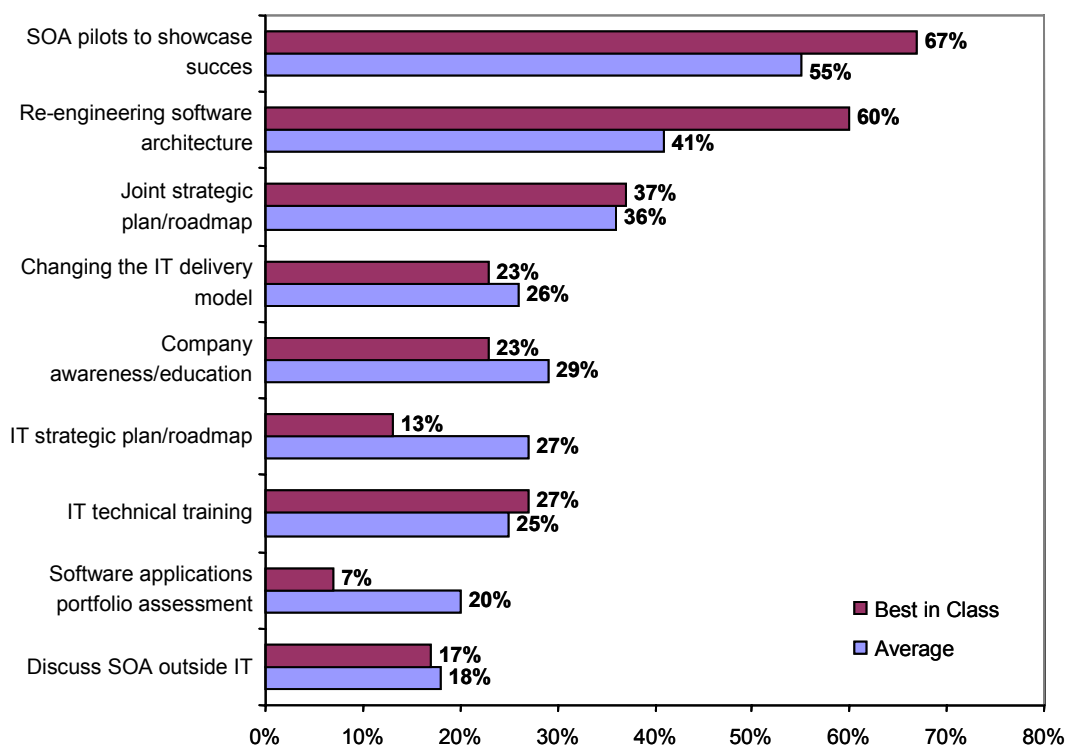
Challenges	% Selected	Responses to Challenges	% Selected
1. Limited visibility for SOA value	41%	1. Small pilot to show success	54%
2. Limited in-house IT skills	34%	2. Systematic planning approach	42%
3. Limited vision/support or understanding from the business side and/or executive leadership	31%	3. Technical education	40%
4. Not enough in-house business analysts versed in SOA	30%	4. Choosing strategic vendors/technologies to work with	33%
5. Change management around people or existing processes	30%	5. Business education	31%
6. Balancing what comes first: services orientation or re-architecting IT	29%	6. An investment-led approach to SOA implementations	24%
7. Lack of best practices or lessons learned	28%	7. Choosing an SOA maturity model as roadmap	16%
8. Too much complexity	22%	8. Investing in SOA complexity management capabilities	14%

Source: [AberdeenGroup](#), December 2005

Experience with SOA Projects Matters

Fifty-five percent of all companies surveyed — and 67% of best-in-class firms — have started SOA projects to showcase early success (Figure 4). Gaining experience in depth and breadth of SOA implementations is a key factor for promoting SOA growth within the IT organization. The reality is that business value will not be achieved before IT value. Therefore, CIOs are encouraged to fully understand how SOA is benefiting their organizations and the resulting metrics for success. A further analysis reveals that 47% of BIC respondents have done more than two projects versus 26% for all respondents in the survey.

Figure 4: Top Three Strategic Actions or Plans to Address/Implement SOA



Source: [AberdeenGroup](#), December 2005

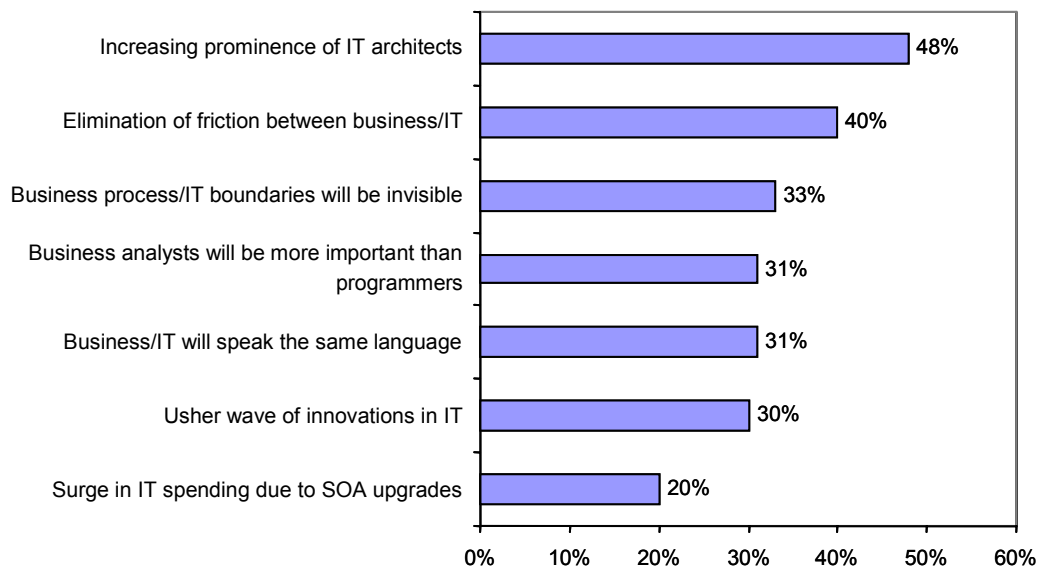
Retooling IT Skills: Business Analysts Are in Demand

Best-in-class companies have a slightly different view on SOA challenges, reflecting their IT maturity level. First, only 33% of BIC companies indicated that limited visibility for SOA value was a challenge (versus 41% for all respondents). Second, 57% of BIC companies cited “not enough in-house business analysts versed in SOA” as a key challenge, in addition to limited in-house IT skills (37% for BIC; 32% for all respondents). This is a true reflection of the impact SOA is having on the IT organization: 48% of respondents believe IT architects will become increasingly prominent, and 31% predict business analysts will become more important than programmers (Figure 5).

However, Aberdeen expects a surge in IT spending in from 2006 to 2008 due to pressures to upgrade a significant percentage of the IT infrastructure to SOAs. The battle over the SOA budget is real because SOA requires an upfront cost investment. The good news is that the ROI is easily achievable.



Figure 5: Likely Scenarios in Three Years as a Result of SOA Adoption



Source: [AberdeenGroup](#), December 2005

Chapter Three: Implications & Analysis

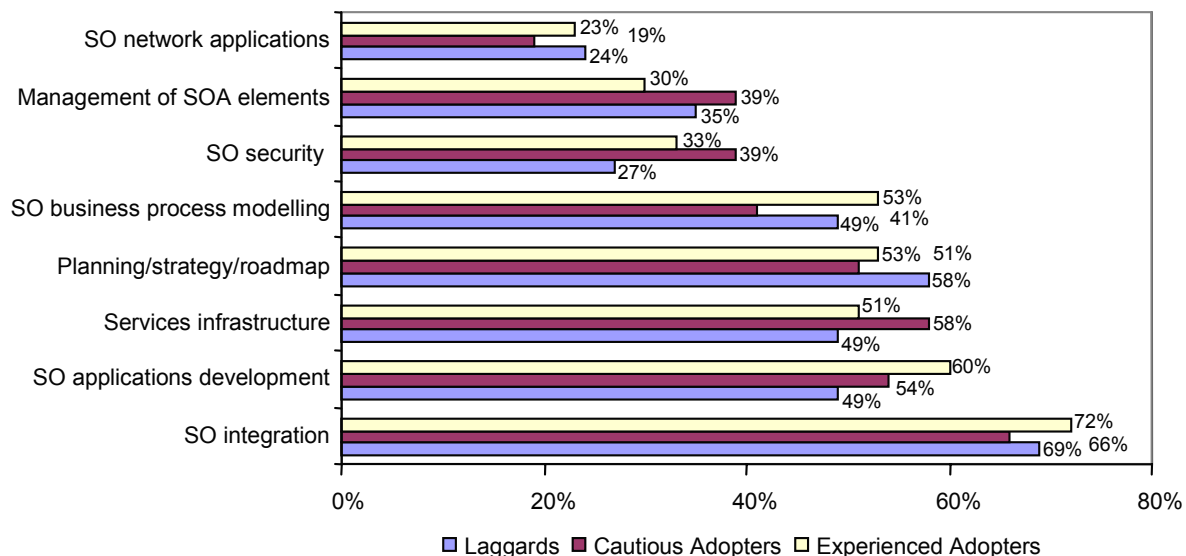
Key Takeaways

- Experienced adopters are focusing on integration, application development, and business process modeling.
- Experienced adopters quantify the value of SOA according to speed of deployment, ease of integration, and faster customization.
- Even best-in-class companies are not taking strategic approaches to SOA planning.
- Best-in-class companies show a much higher rate of adoption in composite applications deployment.

Survey respondents fell into one of three categories – *Laggards*, *Cautious Adopters*, or *Experienced Adopters* — based on their length of experience with SOA projects:

- **Laggards** have had less than 12 months of SOA experience;
- **Cautious adopters** have had between 12 and 24 months of experience; and
- **Experienced adopters** have had more than 24 months of experience.

Figure 6: Areas of SOA Being Implemented



Source: [AberdeenGroup](#), December 2005

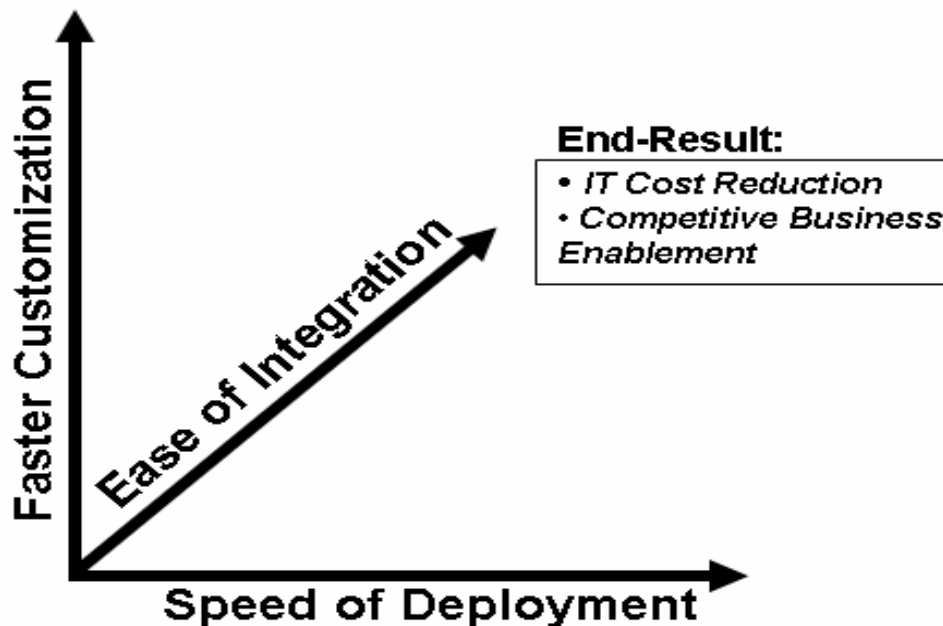
Figure 6 shows that experienced adopters have taken the lead in three key areas of SOA implementation: 1) integration, 2) application development, and 3) business process modeling. This is an accurate reflection of how companies can make a difference in their SOA undertakings.



How to Quantify SOA's Value

Quantifying the results from SOA projects is a key requirement for proving SOA's value and securing further commitments on expanding its footprint. Follow-up interviews with a number of advanced SOA users allowed us to categorize measurable benefits into three categories (Figure 7):

Figure 7: The Three Dimensions of SOA's Value



Source: [AberdeenGroup](#), December 2005

- **Speed of deployment.** One manager of a high-volume application development center cited 40% faster implementations over those of competitors. For this respondent, software development time did not fall with SOA. It was the same, but the overall deployment period was drastically shortened due to reductions in the QA, build, and test cycles. Overall, this resulted in 40% faster deployment.
- **Ease of integration.** SOA forces users to separate application components into logical services. There is less reliance on application programming interfaces for integration purposes. With an SOA approach, you don't need to get into the application's complexity to integrate. Since you're dealing with the encapsulated services, the bugs are not as critical because you're not breaking up the application.
- **Faster customization.** SOA makes it easier to customize or upgrade applications in a step-wise fashion. The concept of live changes during run-time gets closer to reality because it's an integral part of the design requirements. One respondent from a large bank saw a significant outcome after implementing an entire customer-facing web application comprised of 120 services. The bank is now able to

roll out four to six major releases of the application per year instead of two or three before an SOA, so this has become a competitive differentiator.

Key SOA Technologies

Table 3 represents a suggested hierarchy for the various SOA technology pieces. What seems to be lacking is a more widespread adoption of a rigorous strategy planning process for companies involved in SOA. This shows that for half of the respondents, SOA is not a strategic initiative. Even 34% of BIC companies are lagging in taking a strategic approach to SOA.

Table 3: Key SOA Technologies and Current Adoption

A Functional Hierarchy of SOA Elements	% Selected
1. Strategy Planning & Roadmap	50%
2. SOA Business Process Modeling and Design	45%
3. SOA Integration	63%
4. Development of SOA Applications and Services	51%
5. Management of SOA Infrastructure and Services	30%
6. SOA Security	31%
7. SO Network Architecture	21%

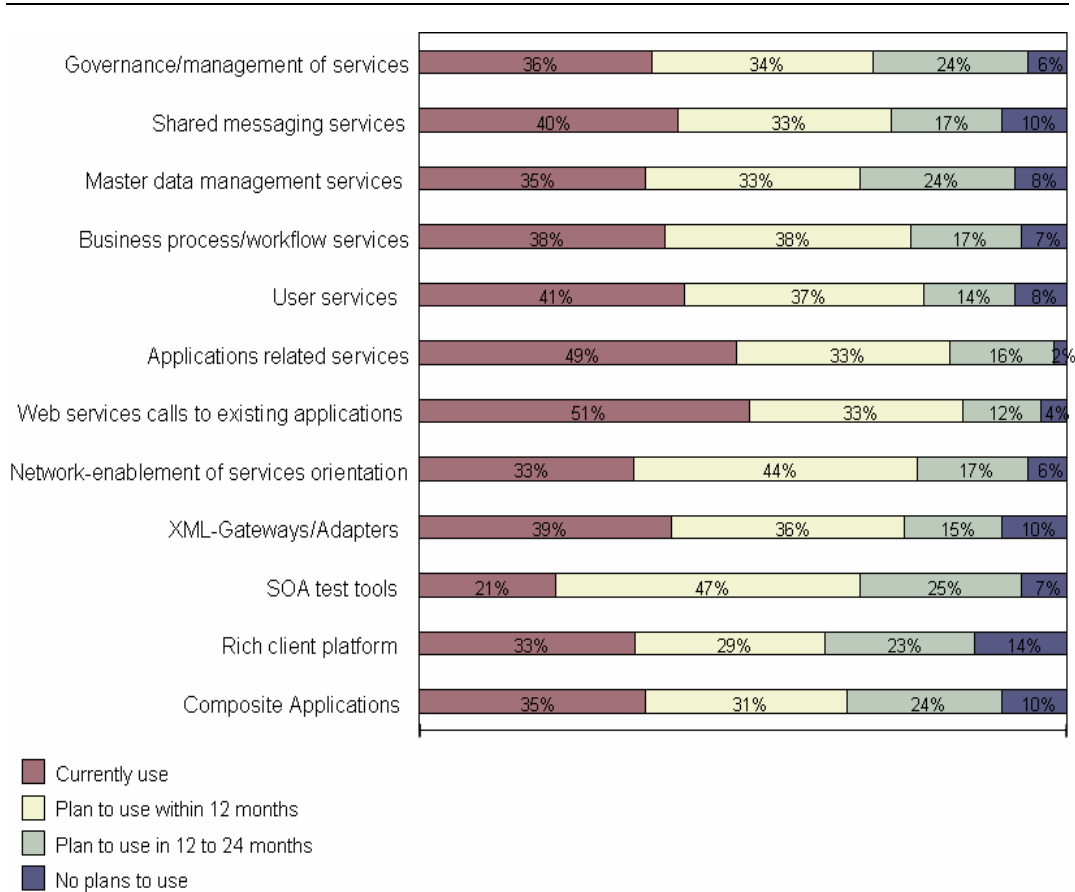
Source: [AberdeenGroup](#), December 2005

Adoption Rates Vary by SOA Technology

Figure 8 shows how organizations are likely to adopt the various SOA technologies over the next 24 months. It's no surprise that the locus of SOA attention seems to revolve around applications. Software applications are indeed the first-candidate technologies that offer the most compelling path to a migration toward an SOA methodology. With composite applications, BIC companies show a higher rate of adoption, 46% currently, versus 29% for all respondents, and 35% over the next 12 months, versus 23% for all respondents.



Figure 8: SOA Technology Components Use



Source: [AberdeenGroup](#), December 2005

Pressures, Actions, Capabilities, Enablers (PACE)

We have analyzed the relationship between the pressures companies identify, the actions they take, and their subsequent competitive performance. Companies interested in SOA should examine their prioritized PACE selections and determine whether they can gain valuable perspectives from Table 4:

Table 4: PACE (Pressures, Actions, Capabilities, Enablers)

Priorities	Prioritized Pressures	Prioritized Actions	Prioritized Capabilities	Prioritized Enablers
1	Managing IT complexity	Re-architecting IT software infrastructure by coupling IT services to business services	Business process integration that enables speed of business changes, including cross-process orchestration and manipulation, and enables event-driven decision making	* Services registry * BPM engine * SOA infrastructure monitoring
2	Re-use of applications via web services	Break up monolithic apps into services building blocks with well defined contracts and policies	Composite applications development Modernization or surrounding of legacy applications with SOA	* WSDL * XML wrappers * Services registry
3	Speed of IT implementations	Demand SOA as a standard for all software development projects	SOBA (service-oriented business applications) is the end goal Interim goal is web services	* SOA development framework such as .NET or J2EE * WS-* standards * SOA test tools
4	Managing IT integration costs	Increased focus on services and process integration, less on APIs	Universal exchange between applications data/logic, XML, - messages, services, rules, policies and external data sources	* SOA ESB (enterprise service bus) * BPM engine
5	SOA security	Consider increased security at the edge of the network or at application-tier level	Securing SOA infrastructure wherever it expands	* SAML, WS-* * Federated identity * XML gateways
6	SOA network architecture	Take a holistic view of SOA by including the role of XML-appliances and network-application processing	XML traffic, networked-application processing as a way to optimize the entire IT infrastructure	Specialized network appliances as part of SOA infrastructure Software that integrates the SOA network edge into the data center

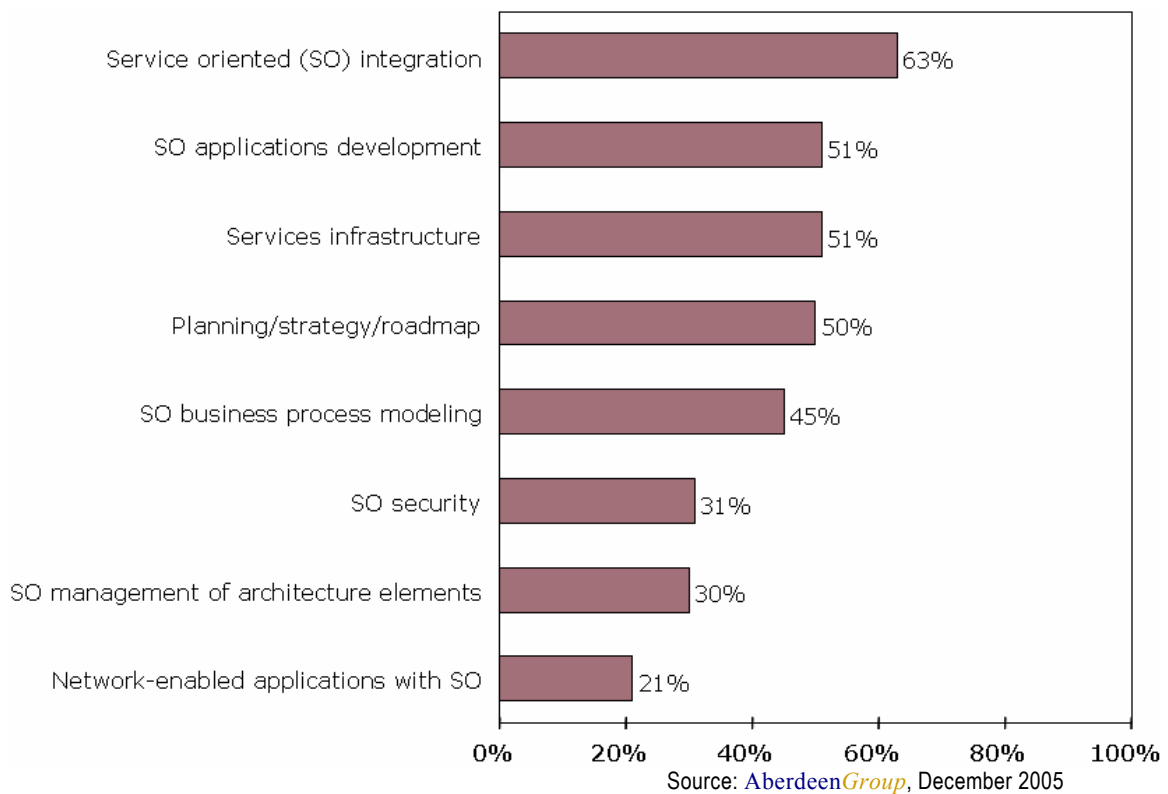
Source: [AberdeenGroup](#), December 2005



Areas of SOA Being Implemented

Figure 9 shows that it's no surprise that integration ranks as the highest SOA activity area within organizations. With the increasing heterogeneity of IT environments, IT integration costs had seen increases in complexity and cost. SOA will help better manage this complexity and lower integration costs due to the standardization aspects of how services interact with each other and what governs such interactions. This is an area of great promise for SOA.

Figure 9: Areas of SOA Being Implemented or Planning to be Implemented



Chapter Four: Recommendations for Action

Key Takeaways

- Best in Class companies have found that SOA adoption lowers IT maintenance costs and enables more spending on innovation.
- Changes within the IT organization are required to mirror the service-delivery aspects of SOA.
- Running IT like a service requires a full business orientation, but SOA is a great catalyst for achieving it.

Cost savings, closer alignment with the business, unparalleled responsiveness, predictable delivery, and ultimately customer satisfaction benefits await all IT organizations that are committed to allowing SOA to permeate their operations. Timid SOA undertakings lower IT's credibility as an agent of change for the business and do not force the IT organization to change its mode of delivery accordingly. If SOA is the exception rather than the norm, there will be a constant battle and many contrasts between traditional and progressive IT undertakings.

Ultimately, SOA is what will tilt the IT budget toward more spending on innovation and will enable new business capabilities, versus a traditional orientation in which maintenance and support costs consume most of the budget. In fact, the IT budget as a percentage of total revenue spent on innovations was 29.6% for best-in-class companies with more than two SOA projects under their belts, versus 18.5% for all survey respondents. And, software maintenance costs as a percentage of IT budgets stood at 12.4% for BIC companies with SOA experience versus 27.3% for all participants.

Recommended Actions for all Companies

SOA is, first and foremost, a technology enabler for the IT organization. But it's also a catalyst of change for how IT delivers its value to the business. CIOs are encouraged to focus on the following capabilities to begin effecting change within their organizations:

- **Take a strategic approach to SOA planning.** Tactical SOA projects are not the only way to proceed. Multi-phased strategic projects may provide a bigger bang for raising the visibility of SOA within the organization at higher levels. CIOs must demand increased budget commitment to bring SOA's promise to fruition. Now is the time to take a holistic and strategic approach for SOA planning, which will include a roadmap, transition plans, and metrics for quantifying SOA's value and results.
- **Invest in an SOA competency center.** Just like the novelty of other innovations, such as the Internet, client-server, and Six Sigma when they were introduced, SOA requires a concentration of knowledge and best practices that's necessary to achieve effective deployments. Lessons learned from any SOA activity are best exploited when there is a centralized approach to its management.



- ***Bring the network architects and network operations into SOA.*** SOA is expansive and the networking experts, especially in large, multi-national organizations, will challenge its turf. It's better to be proactive and agree on the several options are available for bringing the network into the realm of SOA.
- ***Give prominence to a Senior SOA Architect.*** Some progressive organizations are creating a new position, such as a Senior SOA Architect, that will develop the SOA blueprint and roadmap. This is an important position, but it must not be confused with SOA project management. The SOA architect is responsible for the overall planning, strategy and, choice of the various architecture elements, but not necessarily for delivering SOA deployments.
- ***Redesign the business analyst's role.*** Long considered a trivial and not highly visible function, the business analyst takes a new role with SOA due to the recent increase in business process management (BPM) activity. Undeniably, there is some overlap and co-existence between SOA and BPM. Furthermore, as attention shifts to cross-process manipulation and orchestration, the SOA business analyst will become a very important hybrid between IT and the business.
- ***Master the new SOA technologies.*** Aside from trying to assimilate the several SOA product introductions from vendors, there is a handful of key SOA technology pieces that the IT organization must understand well. These include: the enterprise service bus, the services registry, policy governance and management, SOA infrastructure activity monitoring, business process management engines, master data management and semantics building, and Web Services/XML security management.
- ***Develop SOA metrics.*** There is nothing better for showcasing value than the measurement and documentation of specific outcomes. These could range from the obvious — cost, productivity, and time-to-market metrics —to more SOA-specific measurements, such as re-use factor for a Web service, degree of services granularity (from fine to coarse), time to change a governance or usage policy, usage statistics for a given service, value of a service, and linkage of IT service to business service value.

Whether a company is trying to gradually move its SOA practices from Laggard to Industry Average, or Industry Average to Best in Class, the following actions will help spur the necessary performance improvements:

Laggard Steps to Success

Conduct company-wide awareness and education on the value of SOA and its various implementation approaches.

There is nothing better than producing several SOA experts within the organization. SOA's success will not propagate through a single entity or single project.

Get more support and understanding from business and executive leadership.

If you haven't had the luxury of showing internal success yet, there is already a wide body of external successes and best practices with SOA in the marketplace.

Focus on identifying those related to your business case so you can increase the number of SOA related projects.

Industry Norm Step to Success

Adopt the “recommended actions for all companies” (Page 17).

Best in Class Next Steps

Conduct a company-wide software applications portfolio assessment.

After having secured a handful of SOA-related projects and acquiring experience with SOA, fully assess all remaining legacy and existing applications and develop an SOA migration path for each.

Get more support and understanding from the business side and executive leadership.

Armed with the early results and success metrics from completed SOA projects, it's time to raise the visibility for SOA's value within the business levels of the organization. Start waging the battle for increasing the SOA budget.

Start implementing meaningful changes to the IT organization.

Ultimately, SOA is enabling the IT organization to deliver its value as a service. The human resources side has to accompany this new orientation. Although some IT organizations have already implemented internal account management and shared success metrics with the business, SOA ups the ante even more and demands creation of a new hybrid department that sits between the IT we know today and the business unit/operations. This new hybrid has the required skills and capabilities to create, manipulate, design, orchestrate, and mediate a variety of business-level services. If SOA delivers on its promise, all of IT might look like this hybrid in the not-too-distant future.

Let SOA Provide Better IT Predictability

Ultimately, the promise of SOA is about giving IT a higher level of predictability in delivering IT, from technical capabilities to business value. Running IT like a service, adopting open service architecture, and linking business to IT organically via a business services orientation will help the IT organization achieve SO-IT.



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Suite 700
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<http://www.softwareagusa.com>
customerfirst@softwareagusa.com

webMethods, Inc.

3877 Fairfax Ridge Road
South Tower
Fairfax, VA 22030
703 460 2500

Fax: 703 460 2599
<http://www.webMethods.com>



Author Profile

**William Mougayar,
Vice President and Service Director
Information Technology Research
AberdeenGroup, Inc.**

William Mougayar leads Aberdeen's CIO Strategic Agenda research practice, which weaves business strategy and technology operations issues facing CIOs in large and mid-size companies amid the constraints of organizational culture, human resources, budgets, project management, and vendor relations. His current research covers all aspects of SOA, services-oriented network architecture and the next generation of ERP systems. The technology agenda includes all aspects of running a cost-competitive and secure IT infrastructure, delivering robust applications, and shaping a flexible information architecture.

Mougayar has 24 years of strategic, operational, and leadership experience in the IT industry. In the past 10 years, he has consulted and lectured to the world's largest corporations and their CIOs. Previously, he spent 14 years at Hewlett-Packard and held a variety of positions in sales, marketing, and consulting management for enterprise systems in the healthcare, financial services, telecommunications, oil & gas, and telecommunications industries. He also led the client-server, open systems and executive/CIO programs at HP Canada.



Appendix A: Research Methodology

In November 2005, **AberdeenGroup**, with its survey partner, *Accela Communications*, conducted a broad survey of 284 companies involved in SOA technologies and examined their usage patterns, strategies, experiences, intentions, and results. The hypothesis that launched this report is that SOA was fundamentally changing the IT delivery model, the IT organization, and its relationship with the business. While SOA was widely perceived to belong to the technical realm, business managers had to understand the SOA concept better to crack through the barriers of adoption and change.

Responding IT and line-of-business executives, as well as high-level IT contributors, completed an online survey that included questions designed to determine the following:

- Can a better business understanding of SOA lead to more clarity into the evolution of SOA and allow it to reach its full potential in three to five years?
- Can we broaden the discussions outside of SOA and into SO-IT?
- Is IT taking enough leadership to push through the services-oriented agenda, or is it not taking enough risks, and therefore not pushing far enough for breakthroughs?
- How is SOA bringing IT and business together?
- What SOA-created behaviors should IT and its business partners adopt?

Aberdeen supplemented this online survey effort with telephone and e-mail interviews with select survey respondents from the U.S., Europe, and Asia-Pacific countries, gathering additional insights directly from leading practitioners relating to their direct experience in managing or leading SOA projects for their enterprises.

The study aimed to identify the state of SOA adoption and resulting action items necessary to provide a framework by which readers could assess their own SOA capabilities.

Responding enterprises included the following:

- **Job title/function:** The research sample included respondents within the following functional areas: information technology (57%), sales and marketing (17%), business process management (7%). Respondent positions: 17% from executive/senior positions (C-level or SVP); 11% from CIO/IT management; 19%, directors; 22%, managers; and 11%, staff. Within IT, 29% were in general management, 24% in strategy/planning, 14% in software/services architecture, 8% in software development, 5% in infrastructure and 5% in enterprise applications.
- **Industry:** The research sample included respondents from a variety of industries. Discrete manufacturing companies totaled 41% of the sample, followed by services-based industries (24%), consumer-based (19%), public sector (9%) and process manufacturing (4%). Less than half (11% of total) of the services industries respondents consisted of financial services institutions.

- **Geography:** There was a good mix of global distribution from survey respondents. Close to half (49%) were from North America, 30% from EMEA, 16% from Asia-Pacific, and 6% from South America.
- **Company size:** About 30% of respondents were from large enterprises (annual revenues of more than \$1 billion); 28% were from mid-size enterprises (annual revenues between \$50 million and \$1 billion); and 41% were from small businesses (\$50 million or less).

Solution providers recognized as sponsors of this report were solicited after the fact and had no substantive influence on the direction of *The SOA in IT Benchmark Report*. Their sponsorship has made it possible for [AberdeenGroup](#) and *Accela Communications* to make these findings available to readers at no charge.

Table 5: Relationship between PACE and Competitive Framework

PACE and Competitive Framework How They Interact
Aberdeen research indicates that companies that identify the most impactful pressures and take the most transformational and effective actions are most likely to achieve superior performance. The level of competitive performance that a company achieves is strongly determined by the PACE choices that they make and how well they execute.

Source: [AberdeenGroup](#), December 2005



Appendix B: **Related Aberdeen Research & Tools**

Related Aberdeen research that forms a companion or reference to this report includes:

- [*The ESB in the Land of SOA*](#) (December 2005)
- [*The Service-Oriented Architecture in the Supply Chain Benchmark Report*](#) (September 2005)

Information on these and any other Aberdeen publications can be found at www.Aberdeen.com.

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AberdeenGroup, Inc.
260 Franklin Street
Boston, Massachusetts
02110-3112
USA

Telephone: 617 723 7890
Fax: 617 723 7897
www.aberdeen.com

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